DATA CLEANING & PREPARATION

Data Analytics – Project 1

DecodeLabs Industrial Training Kit | Batch 2026

Project	Data Analytics Project 1
Focus	Data Cleaning & Preparation
Dataset	Dataset for Data Analytics(1).xlsx
Worksheet	Sheet1
Records	1,200
Columns	14
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Purpose

This report documents the data-quality assessment and cleaning preparation performed on the supplied order dataset. The assessment focuses on the Project 1 requirements: identifying missing values, checking duplicates, validating unique IDs, and checking data formats before further analysis.

Report status: Dataset assessment completed. Cleaning actions are documented separately from validation results so that no unperformed change is presented as completed.

1. Dataset Overview

The supplied Excel workbook contains one worksheet, Sheet1, with 1,200 records and 14 columns. The dataset represents order-level information including customer, product, payment, shipping, tracking, coupon, referral, and pricing fields.

Column	Data Type	Description
OrderID	Text	Order identifier
Date	DateTime	Order date
CustomerID	Text	Customer identifier
Product	Text	Product name
Quantity	Integer	Number of units ordered
UnitPrice	Decimal	Price per unit
ShippingAddress	Text	Shipping address
PaymentMethod	Text	Payment method
OrderStatus	Text	Order status
TrackingNumber	Text	Shipment tracking number
ItemsInCart	Integer	Number of items in cart
CouponCode	Text	Coupon/promotion code
ReferralSource	Text	Customer referral/source
TotalPrice	Decimal	Total order value

2. Initial Data Quality Assessment

Check	Actual Result
Total records	1,200
Total columns	14
Missing values – CouponCode	309
Missing values – all other columns	0
Duplicate complete rows	0
Total OrderIDs	1,200
Unique OrderIDs	1,200
Duplicate OrderIDs	0
Date minimum	2023-01-01
Date maximum	2025-06-30
Date missing values	0

Interpretation: The primary missing-data issue is in CouponCode. The supplied dataset contains no complete duplicate rows and no duplicate OrderIDs. The Date field is populated throughout the dataset and is stored as a datetime field.

3. Missing Value Analysis

Missing-value analysis was performed for every column. Only CouponCode contains missing values: 309 of 1,200 records. All other 13 columns contain zero missing values.

Column	Missing	Non-Missing	Assessment
CouponCode	309	891	Missing values identified
All other columns	0	1,200 each	No missing values identified

Recommended Cleaning Action

For analysis, the 309 missing CouponCode values can be replaced with the explicit category **No Coupon**. This preserves the distinction between an absent coupon and an unknown value while making the column easier to group and analyze. This replacement should be recorded as a cleaning action only after it has actually been applied to the cleaned workbook.

4. Duplicate Record Analysis

The workbook was checked for exact duplicate rows. The result is 0 duplicate rows, so no complete-record deletion is required.

Duplicate Check	Result	Conclusion
Complete duplicate rows	0	No duplicate rows found
OrderID total	1,200	All order records checked
Unique OrderID	1,200	All OrderIDs are unique
Duplicate OrderID	0	No duplicate OrderIDs found

OrderID Verification Method

In Excel, the uniqueness of each OrderID can be verified with the following formula:

Excel formula
=COUNTIF($A:$A,A2)

A result of 1 indicates that the OrderID appears once in the OrderID column. A result greater than 1 indicates a duplicate OrderID. In this dataset, the observed duplicate count is 0.

Important: repeated CustomerIDs should not automatically be treated as duplicate orders, because one customer can legitimately place multiple orders.

5. Date Format Validation

The Date column contains 1,200 non-missing values and is stored as a datetime field in the supplied Excel workbook. The observed date range is from 2023-01-01 through 2025-06-30.

Validation	Result
Date data type	datetime
Missing dates	0
Earliest date	2023-01-01
Latest date	2025-06-30
Invalid/unreadable dates	0 observed

For consistent presentation and downstream analysis, dates can be displayed in a standard format such as **YYYY-MM-DD**. Because the source values are already stored as valid datetime values, this is primarily a formatting/standardization step rather than a data-repair operation.

6. Numeric Data Validation

The numeric fields were checked for their stored data types. Quantity and ItemsInCart are integer fields, while UnitPrice and TotalPrice are decimal/numeric fields.

Field	Stored Type	Validation
Quantity	Integer	Numeric
UnitPrice	Decimal	Numeric
ItemsInCart	Integer	Numeric
TotalPrice	Decimal	Numeric

These fields are therefore suitable for numerical calculations and aggregation, subject to normal business logic checks in later analysis.

7. Cleaning Actions and Verification Plan

Issue	Observed	Required/Recommended Action	Final Verification
Missing CouponCode	309	Replace with No Coupon	Missing CouponCode = 0
Duplicate rows	0	No deletion required	Duplicate rows = 0
Duplicate OrderID	0	No correction required	Duplicate OrderID = 0
Date values	0 missing	Standardize display if needed	Valid date format
Numeric fields	Numeric types	No type conversion required	Numeric types verified

Note on completion: The dataset assessment confirms the current source-data condition. The CouponCode replacement should be applied in the separate cleaned workbook before claiming the cleaned dataset has zero missing CouponCode values.

8. Before vs. After Summary

The table below separates the observed source condition from the intended cleaned-state verification. This avoids presenting a recommended action as completed before it is actually performed.

Quality Check	Source Dataset	Target After Cleaning
Missing CouponCode	309	0
Duplicate complete rows	0	0
Duplicate OrderID	0	0
Missing Date	0	0
Invalid Date	0 observed	0
Numeric field types	Verified numeric	Verified numeric

9. Final Verification Checklist

Verification Item	Status
All 1,200 records reviewed	Completed
All 14 columns assessed	Completed
Missing values identified	Completed
Duplicate rows checked	Completed
OrderID uniqueness checked	Completed
Date field validated	Completed
Numeric field types validated	Completed
CouponCode missing values replaced	To be applied in cleaned workbook
Final cleaned workbook rechecked	Required after cleaning

10. Conclusion

The supplied dataset is structurally clean in several important areas: it contains no complete duplicate rows, all 1,200 OrderIDs are unique, the Date field has no missing values, and the principal numeric fields are stored using numeric data types. The main identified data-quality issue is 309 missing CouponCode values. Replacing those values with an explicit No Coupon category is the recommended cleaning action. After applying that change, the cleaned workbook should be rechecked and the final verification results recorded.

11. Tools Used

Microsoft Excel; COUNTIF; Conditional Formatting; Remove Duplicates; Sorting/Filtering; Data Type Validation.

12. Project Reference

This report is prepared for Data Analytics Project 1: Data Cleaning & Preparation in the DecodeLabs Industrial Training Kit, Batch 2026.

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